

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2024

Wednesday 22 May, 09.30 – 12.30

There are TEN questions in this paper.

Answer ALL Part A questions and TWO Part B questions.

Part A attracts 40% of the total marks. Each question in Part A has the same weight. Each question in Part B attracts 30% of the total marks.

For multipart questions, the weight assigned to each part is indicated in brackets.

Candidates may use a calculator as specified by the Department of Economics.

Do **not** turn over until told that you may do so.

Part A

1. In an exchange economy Andrea and Bee each have Cobb-Douglas utility functions over goods x and y . Andrea's consumption bundle is $\{x_a, y_a\}$ and Bee's is $\{x_b, y_b\}$. The utility functions are, for Andrea first and Bee second,

$$u_a(x_a, y_a) = 2(\ln x_a + \ln y_a)$$

$$u_b(x_b, y_b) = \ln x_b + \ln y_b.$$

Andrea has an endowment of 18 units of good x and 6 units of good y . Bee's endowment is 6 units of good x and 18 units of good y .

- (a) Let the price of good y be 1, so the relative price of good x is p . Show that the competitive equilibrium relative price of x is $p = 1$, and that in the equilibrium allocation Andrea consumes 12 units of each good and Bee consumes 12 units of each good. State the conditions for an allocation to be Pareto efficient, and show that the allocation $(12, 12; 12, 12)$ satisfies them.

[55%]

- (b) A social planner can transfer good y between the two agents to alter the initial endowments. Good x cannot be transferred. Explain why such a transfer does not change the equilibrium relative price.

[15%]

- (c) If the planner is utilitarian, and so maximizes the sum of the utilities, how much of good y should be transferred (starting from the initial endowments), and who should receive the transfer?

[30%]

2. Alicia and Bob have quasi-linear utility functions over the quantity of a public good, q , and the money spent on a private good, x_a for Alicia and x_b for Bob. Alicia's utility function is

$$U_a(q, x_a) = 2q - q^2 + x_a.$$

Bob's utility function is

$$U_b(q, x_b) = q - \frac{1}{2}q^2 + x_b.$$

The cost of buying a unit of the public good is c , which satisfies $0 < c < 1$. Assume that Alicia and Bob have incomes, m_a and m_b respectively, that are large enough that each will consume a positive amount of the private good, so $x_a > 0$ and $x_b > 0$.

- (a) Suppose that Alicia and Bob independently choose how much of the public good to buy. How much would they buy if the other person buys nothing? Show that the Nash equilibrium with private provision of the public good has Alicia buying $1 - \frac{c}{2}$ and Bob buying zero of the public good. Explain.

[50%]

- (b) What, in general, is the Samuelson condition for public goods provision? How much of the public good is provided when the Samuelson condition holds in this economy?

[20%]

- (c) Professor Lindahl knows that Alicia and Bob always tell the truth. He suggests to the government that (i) each agent should be asked to report their marginal utility function for the public good, (ii) the Samuelson rule should determine the quantity of the public good, and (iii) each agent should pay a price per unit of the public good equal to their marginal utility at the quantity provided. (Equivalently each agent could be asked to report their demand function for the public good, with the government choosing the individual prices so that the demands are equal and the sum of the prices equals c .)

Discuss the reasoning behind Professor Lindahl's proposal, and whether it is likely to be a good scheme in practice.

[30%]

3. Row and Column play the following 2x2 strategic-form game. (Row picks the row, either T or B; Column picks the column, either L or R.) Row's payoff is written first in each cell. Both players know the value of x , to be specified below.

	L	R
T	2, 1	1, x
B	3, 1	0, 2

- (a) Assume $0 < x < 1$. Find all Nash equilibria of the game in which Row and Column choose simultaneously. Compute the players' equilibrium payoffs. How do the players' equilibrium strategies and payoffs depend on x ? Provide an explanation (intuition) for your findings. [40%]
- (b) Assume $0 < x < 1$. Explain whether Row or Column would weakly prefer to move first (committing to an action that is publicly observed) rather than choosing simultaneously. [25%]
- (c) Find an example of a two-player strategic-form game with a unique Nash equilibrium (in pure or mixed strategies), in which both players are strictly worse off moving first, provided their move is observed by the other player. [15%]
- (d) Assume instead $x = 1$. Find all Nash equilibria of the game with simultaneous moves. [20%]

4. Ava is an expected utility maximiser with monetary utility $u(x)$, where u is strictly increasing. Consider the lotteries $L = [1 - p, p; 0, y]$ and $L' = [1 - q, q; 0, z]$, where the probabilities p and q belong to $(0, 1)$ and the monetary prizes satisfy $z > y > 0$.

(a) Assume first that $q \geq p$. Does any of the lotteries first-order stochastically dominate the other?

[30%]

(b) Show that if Ava prefers L to L' , it must be that $p > q$. If we assume $u(0) = 0$ and $u(z) = 1$, what are the possible values of $u(y)$ in this case?

[30%]

(c) Prove formally that an expected utility maximiser with monetary utility $v(x) = \alpha + \beta u(x)$, where $\beta > 0$, has the same risk preferences as Ava does. Use this to defend the idea that the assumptions $u(0) = 0$ and $u(z) = 1$ above are harmless.

[40%]

Part B

5. Should microeconomic policy be guided by the principle that a policy should be implemented if the winners can compensate the losers and still be better off? If so, explain why. If not, how should microeconomic policy be determined?
6. For *either* emissions of greenhouse gases *or* traffic congestion in cities, discuss the relative merits of the following methods of addressing the externalities: (i) establishing and enforcing property rights, (ii) Pigouvian taxes, (iii) tradeable permits (cap-and-trade) schemes and (iv) direct regulation (which limits by law emissions from each source, or limits when a vehicle may enter a city).
7. Discuss, in the context of one or more suitable models of imperfect competition, under what conditions firms would prefer to merge rather than collude (and vice versa), and whether (or when) a regulator may prefer one outcome or the other (merger versus collusion).
8. A sordid club offers the following opportunity: you can win an obscenely huge prize or you can be dispossessed of everything you own. The club proudly claims that they can convince any expected utility maximiser to play by personalising the probability of winning in the game. The club also argues that, given a probability and a prize, any expected utility maximiser who wants to play the game would also want to put their name on a list from which one name would be randomly chosen to participate in the game. Using these two ideas, state formally the properties of continuity and independence and prove formally that expected utility satisfies them. Given a probability and a prize, what type of expected utility maximisers would you expect to play such a game?
9. “When information is symmetric, adverse selection and moral hazard are not relevant in the economic situation at hand; when information is asymmetric, adverse selection and moral hazard are equally relevant to understand the economic situation at hand.” Discuss.

10. A risk neutral firm considers hiring an agent for an important project. The project may end up in high (H), medium (M), or low (L) success, resulting in profits of 400 pounds, 280 pounds and 100 pounds, respectively. The probability of each type of success depends on whether the agent puts in high (h) or low (l) effort, as follows:

- If effort is high ($e_h = 1$), the corresponding probabilities are $\frac{1}{2}$, 0 and $\frac{1}{2}$.
- If effort is low ($e_l = 0$), the corresponding probabilities are $\frac{1}{4}$, $\frac{1}{4}$ and $\frac{1}{2}$.

The candidate is an expected utility maximizer with utility function $u(w, e) = \sqrt{100 + w} - e$. The candidate has a reservation utility of 11 and effort is not observable. The firm will be using contracts of the form (w_H, w_M, w_L) , contingent on the level of success of the project.

- (a) Discuss very briefly why the absence of observability leads to a moral hazard problem. If the firm wants to induce low effort, which contract will be offered? What would be the profits of the firm?

[15%]

- (b) Suppose that the firm wants to induce high effort. Write down the conditions that are needed for the agent to accept the contract and put in high effort.

[15%]

- (c) Suppose that the firm wants to induce high effort and can use any wages in $[-100, +\infty)$. Explain why setting a negative wage of $w_M = -100$ would be a good idea for the firm. Given this, why is a good idea to offer a contract in which $w_H = w_L$? What is then the optimal contract and the profits of the firm?

[20%]

- (d) Suppose that the firm wants to induce high effort but wages cannot be negative by law. Argue that the firm will set $w_M = 0$. Which are the optimal values of w_H and w_L for the firm in this situation?

[20%]

- (e) Comparing the profits of inducing both levels of effort, determine which contract will be offered by the firm if wages cannot be negative by law. Which is the agency cost?

[15%]

- (f) Suppose that the agent is allowed to give up the right to non-negative wages. Explain how this could generate a Pareto superior situation in which both the firm and the agent are strictly better off.

[15%]